

UDS Diagnostic Protocol & Flash Bootloader Validation Report

ISO 14229 Diagnostic Services (0x10, 0x19, 0x27, 0x36) and OTA Rollback

1. Diagnostic Session Control (0x10) Server Response Timing

- Test Case TC-UDS-061: UDS Service 0x10 diagnostic timing test: measured P2 server response time across 100 requests was 18.4 ms

2. DTC Freeze Frame Snapshot Storage

- Test Case TC-UDS-062: DTC snapshot validation test: freeze frame written to NVRAM in 12.8 ms with complete mandatory sensor records

3. UDS Security Access (0x27) Seed-Key Authentication

- Test Case TC-UDS-063: Level 0x01 Reprogramming security access validated with ECDSA-256. Level 0x03 Engineering calibration access successful

4. Flash Bootloader Rollback & Security Findings

- Test Case TC-UDS-064: Corrupted image injected into Bank B; bootloader detected CRC mismatch and initiated rollback in 1.4 seconds
- Finding SEC-065: Section 3.2: Diagnostic port configuration revealed Service 0x23 ReadMemoryByAddress is configured with open access
- Test Case TC-UDS-066: Flash throughput benchmark: measured write speed was 38.5 kB/s due to internal flash block erase wait states